User Interaction in AI-Enhanced Immersive XR Environments: A Systematic Literature Review

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Abstract

Extended reality (XR) technologies, encompassing virtual reality (VR), augmented reality (AR), and mixed reality (MR), are increasingly enhanced by artificial intelligence (AI) to enable adaptive, human-centered immersive environments. These intelligent systems aim to personalize interaction by integrating multimodal sensing, intelligent feedback, and context-aware adaptation. While individual applications have been reported across various domains, including education, healthcare, industry, and cultural heritage, a consolidated understanding of how user interaction is enabled and enhanced remains fragmented. This paper presents a systematic literature review (SLR) of user interaction in immersive XR environments, with a focus on personalization, adaptivity, and intelligent system integration. Following a comprehensive search across three databases (Scopus, ACM, IEEE), which yielded 982 papers, a refined selection of 93 publications was made to conduct an in-depth analysis of cutting-edge research in the field. Guided by four research questions, the review identifies application domains and interaction modalities, examines hardware interfaces and sensor technologies, analyzes the use of AI and non-AI techniques for adaptive user interaction, and synthesizes current limitations and future opportunities. This review highlights a shift from device-centric systems toward multimodal ecosystems, with opportunities in real-time adaptation, multimodal fusion, cross-domain scalability, and ethical personalization. The synthesis provides a structured foundation for advancing intelligent, user-centric XR and informs future research on building responsive, context-aware immersive systems.

Keywords: Extended reality (XR), virtual reality (VR), augmented reality (AR), mixed reality (MR), human-computer interaction (HCI), artificial intelligence (AI),

personalized interaction, multimodal interaction, real-time interaction, immersive environments, sensor-based interaction

1 Introduction

Extended reality (XR), encompassing virtual reality (VR), augmented reality (AR), and mixed reality (MR), is increasingly shaping modern digital ecosystems. Recent advances have moved beyond conventional XR systems that deliver static or uniform experiences, toward AI-enhanced immersive environments that leverage artificial intelligence (AI), machine learning (ML), and multimodal sensing to adapt dynamically to users. These intelligent environments have the potential to transform how individuals learn, collaborate, receive healthcare, and engage with cultural or entertainment content. By tailoring content, interfaces, and feedback to individual needs, such environments aim to achieve deeper immersion, improved accessibility, and more meaningful engagement.

At the heart of this transformation lies the notion of intelligent interaction. In XR systems, interaction is not merely a mechanism to navigate or manipulate virtual spaces but a central element of adaptation itself. The way users communicate with immersive environments, through speech, gestures, gaze, haptic cues, or physiological signals, defines the fluidity, intuitiveness, and responsiveness of the experience. Intelligent interaction, whether powered by AI or enabled through carefully designed non-AI techniques, plays a crucial role in reducing cognitive load, anticipating user intent, and harmonizing the virtual and physical worlds. In this sense, adaptation is twofold: the environment adjusts to the user, and the interaction modalities themselves evolve to better support human needs and contexts.

This centrality of interaction is evident across diverse application domains. In education, adaptive XR can personalize learning pathways; in healthcare, multimodal sensing enables tailored therapeutic interventions; in collaborative work, AI-driven avatars and telepresence systems facilitate richer exchanges; and in cultural or entertainment settings, immersive narratives are made more engaging through real-time responsiveness. Yet, realizing this vision also raises critical challenges related to usability, inclusivity, ethical personalization, and the human-centered design of adaptive systems.

The goal of this systematic literature review (SLR) is to provide a comprehensive investigation of user interaction in human-centered and AI-enhanced immersive XR environments. Specifically, it aims to identify the application domains, purposes, and interaction modalities employed in such environments, examine the hardware interfaces and sensor technologies enabling real-time interactions, and analyze the role of AI and non-AI techniques in supporting adaptive, personalized user experiences. Finally, it explores the current limitations and emerging opportunities in designing adaptive interaction methods, offering insights to guide future research and development of immersive, responsive, and context-aware XR systems; see below for research questions (RQs).

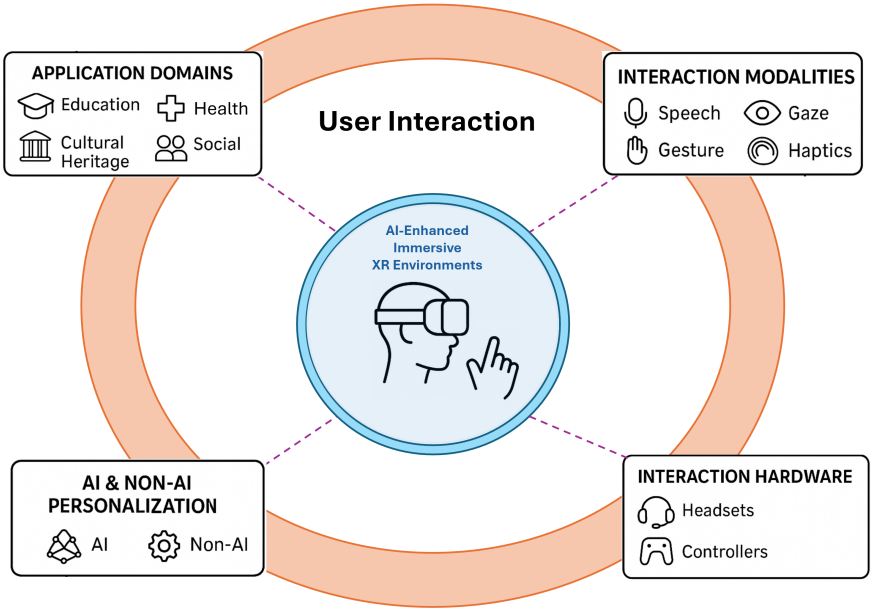


Fig. 1: The scope of our SLR to study user interaction in AI-enhanced immersive XR environments, as also defined by our four research questions.

- **RQ1**: What are the application domains, purposes, and interaction modalities of interaction in human-centered or AI-enhanced immersive XR environments?
- **RQ2**: What hardware interfaces and sensor technologies enable real-time interaction in human-centered or AI-enhanced immersive XR environments?
- **RQ3**: How are AI and non-AI techniques utilized for adaptive, human-centered, or personalized user interaction in human-centered or AI-enhanced immersive XR environments?
- **RQ4**: What are the current limitations and emerging opportunities in designing adaptive interaction methods for human-centered or AI-enhanced immersive XR environments?

It is worthwhile mentioning that, whereas we have placed the constraint on identified immersive XR environments to be intelligent through the use of AI, we have not imposed a restriction on the AI component in terms of user interaction. By systematically synthesizing existing evidence, this SLR aims to deliver a structured account of the state of the art, uncover knowledge gaps, and inform the development of next-generation XR applications that are both intelligent and user-centric. Furthermore, this work contributes a consolidated understanding of how XR interactions are evolving and highlights pathways for developing next-generation immersive environments that are not only technologically advanced but also responsive, ethical, and deeply human-centered (Fig. 1).

2 Related work

In recent years, several surveys and reviews have examined aspects of immersive XR and AI-enhanced environments, though often with differing scopes; also see Tab. 1. Marín-Morales et al. [1] conducted one of the earlier systematic reviews on emotion recognition in VR, focusing on physiological and behavioral measures with head-mounted displays (HMDs). More recent efforts, such as Partarakis and Zabulis [2], have expanded toward human-centered interactions in the metaverse, emphasizing the convergence of XR with AI across multiple domains. In the education sector, Wang and Huang [3] systematically reviewed AI-driven VR/AR systems for adaptive learning, while Huang et al. [4] provided an SLR on smart personalization in immersive XR, highlighting the role of AI/ML in gesture and eye-tracking–based adaptation. Similarly, Lampropoulos [5] focused on intelligent tutoring systems integrated with XR to deliver adaptive learning feedback. Beyond education, Karthikeyan and Aruna [6] reviewed optimization techniques for user interaction and user experience (UX) in XR, covering immersion, presence, and multimodal feedback.

While these reviews collectively highlight the growing role of AI in XR, they are either domain-specific (e.g., education or tutoring systems), technology-specific (e.g., physiological sensing, personalization), or limited to general interaction/UX optimization. None provides a comprehensive, cross-domain analysis of how AI and non-AI techniques are utilized for adaptive, human-centered, and personalized user interaction across immersive XR environments. This gap motivates our study, which systematically maps and analyzes the state of the art in user interaction for AI-enhanced immersive XR.

3 Method

In this study, we adopt the SLR framework from [7] to maintain clarity, rigor, and comprehensiveness in the review process. The approach is guided by well-defined research questions and involves a structured search for identifying relevant published work, applying the pre-defined inclusion (IC) and exclusion (EC) criteria. The objective is to identify and evaluate research related to user interaction in human-centered, AI-based immersive XR environments across various application domains, as well as to gather and analyze the corresponding data.

The subsequent sections detail our data collection process, starting with the development of search strings and execution of searches across three primary databases. This is followed by the presentation of the IC/EC criteria and a description of the multi-stage data filtering procedure conducted collaboratively by all authors in line with the predefined criteria. Finally, we summarize the dataset obtained after two rounds of rigorous screening. Fig. 2 provides a brief overview of the procedure followed in this study.

3.1 Data collection

Once the review aim is defined, we have started our data collection process as follows. The search strings were formulated based on the four proposed RQs, and retrieved

Table 1: Related work, surveys, and reviews.

Reference	Year	Application Domain	Scope	SLR	Articles analyzed
Marín-Morales et al. [1]	2020	Immersion and Emotion Recognition	Physiological and behavioral measures in VR (HMDs)	Yes	144
Partarakis and Zabulis [2]	2024	Interactions in Metaverse	Human-centered XR/AI convergence across multiple domains	No	220
Wang and Huang [3]	2025	AI in Education (Metaverse)	Adaptive learning with AI-driven VR/AR systems	Yes	42
Huang et al. [4]	2025	Personalized Immersive XR	Smart personalization using AI/ML, gesture and eye tracking	Yes	94
Lampropoulos [5]	2025	Intelligent Tutoring in Metaverse	ITS integration with XR for adaptive feedback and learning	Yes	32
Karthikeyan and Aruna [6]	2025	User Interaction and UX in XR	Optimization techniques for immersion, presence, multimodal feedback	No	61
Ours	2025	User Interaction in AI/XR	User Interaction in AI-Enhanced Immersive XR Environments	Yes	99

papers from the Scopus, ACM, and IEEE libraries. The IC/EC criteria were also established in accordance with the review objectives and the SLR guidelines. All authors were actively involved in filtering and evaluating the retrieved publications, ensuring that only studies meeting the specified review objectives and criteria were included in the final analysis.

3.1.1 Search string

Taking into consideration the purpose of our review and the five research questions introduced earlier, the search string was developed as follows:

TITLE-ABS-KEY ((*"virtual reality"* **OR** *"augmented reality"* **OR** *"mixed reality"* **OR** *"extended reality"* **OR** *VR* **OR** *AR* **OR** *MR* **OR** *XR* **OR** *"3D scene"* **OR** *"3D environ*"*) **AND** (*immers** **OR** *"user experience"* **OR** *"user engagement"*) **AND** (*AI* **OR** *"artificial intelligence"* **OR** *ML* **OR** *"machine learning"* **OR** *DL* **OR** *"deep learning"* **OR** *"generative AI"* **OR** *"agent-based"*) **AND** (*personali** **OR** *adapt** **OR** *custom** **OR** *"user-cent*"* **OR** *"human-cent*"* **OR** *"context-aware*"* **OR** *"user-aware*"* **OR** *"affective computing"* **AND**) (*""user interaction"*

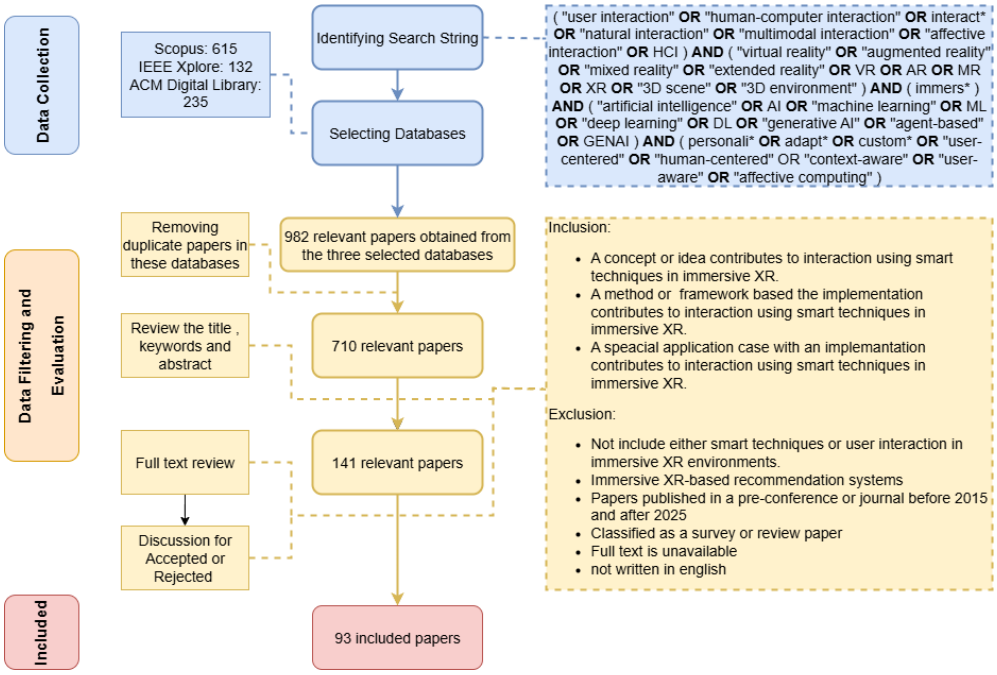


Fig. 2: Our SLR method includes data collection, data filtering, and an evaluation process starting with three different sources (Scopus, ACM Digital Library, and IEEE Xplore).

OR *"human-computer interaction"* **OR** *"interact** **OR** *"natural interaction"* **OR** *"multimodal interaction"* **OR** *"affective interaction"*))

The Search string can be analysed by breaking down with the following components (see also Tab. 2):

- The first substring captures the concepts of XR environments, which form the foundation of our research questions and overall study. To achieve this, the string incorporated terms such as virtual reality, augmented reality, mixed reality, along with their abbreviations (VR, AR, MR), as well as variations like "3D environ"* and "3D scene."
- To capture immersive and engaging experiences, the second component integrates terms like "immers," "user experience,"* and "user engagement."
- The third component addresses the incorporation of AI-related approaches by including keywords such as AI, artificial intelligence, ML, machine learning, DL, deep learning, generative AI, and agent-based.
- The fourth component introduced terms such as "personali," "adapt*," "custom*," "user-cent*," "human-cent*," "context-aware*," "user-aware*,"* and "affective computing" to ensure coverage of human-centered, adaptive, and personalized XR applications.

- Finally, since user interactions are the core of this SLR, the fifth component focuses on user interaction paradigms on established environments by including expressions such as "user interaction," "human-computer interaction," "interact*," "natural interaction," "multimodal interaction," and "affective interaction."

Table 2: Based on our objective of the review, the five components of the search string

Target	Search string component
XR environments	("virtual reality" **OR** "augmented reality" **OR** "mixed reality" **OR** "extended reality" **OR** VR **OR** AR **OR** MR **OR** XR **OR** "3D scene" **OR** "3D environ*")
Immersive Experience	(immers* **OR** "user experience" **OR** "user engagement")
AI-integration	(AI **OR** "artificial intelligence" **OR** ML **OR** "machine learning" **OR** DL **OR** "deep learning" **OR** "generative AI" **OR** "agent-based")
Personalization	(personali* **OR** adapt* **OR** custom* **OR** "user-cent*" **OR** "human-cent*" **OR** "context-aware*" **OR** "user-aware*" **OR** "affective computing")
User Interaction	("user interaction" **OR** "human–computer interaction" **OR** HCI **OR** "interact*" **OR** "natural interaction" **OR** "multimodal interaction" **OR** "affective interaction")

3.1.2 Search procedure

We have chosen three key digital libraries for the search: Scopus, ACM Digital Library, and IEEE Xplore. These sources were selected to provide broad coverage of relevant research across journals, conferences, and workshops, ensuring inclusion of both short and full-length studies. Scopus was selected due to its broad coverage, ensuring that a wide range of relevant studies was captured. In addition, ACM Digital Library and IEEE Xplore, both highly recognized within the field, were also included in the search for their relevance to computing and human–computer interaction research. Although some overlap with Scopus results was expected, incorporating these databases helped minimize the risk of missing potentially important publications.

The database search produced a large number of results across the three sources without the removal of duplicates. The search string was applied to the fields of title, abstract, and keywords in each database. Among the three, Scopus returned the largest set of articles, with 615 records. In comparison, ACM Digital Library contributed 235 papers, while IEEE Xplore produced 132 papers. We further processed the duplication for removal of duplicates; the final count after removal of duplicates is shown in Tab. 3. We concluded our literature retrieval on August 7th, 2025, and subsequently initiated the analysis of the retrieved papers.

3.2 Inclusion/exclusion criteria

A publication was *included* for further analysis if both of the following conditions were met:

Table 3: The number of publications retrieved from Scopus, ACM Digital Library, and IEEE Xplore, based on the search string applied to author keywords, abstract, and document title fields.

Data fields	Database	Number of publications
Abstract |Author Keywords |Document Title	Scopus	372
	ACM	206
	IEEE Xplore	132

- *IC1*: it proposes a concept, idea, or implementation of a personalized, immersive, AI-enhanced XR environment
- *IC2*: it includes an AI or non-AI-based user interaction with the AI-centered immersive XR environment

A publication is *excluded* from further analysis if any of the following criteria were not met:

- *EC1*: it was not published between 2015 and 2025
- *EC2*: it does not have an XR environment that is not smart or personalized and AI-centered
- *EC3*: it does not include any user interaction with the environment
- *EC4*: the full text is not available
- *EC5*: it is not written in English
- *EC6*: it is a survey, review, or SLR paper, in which case it was included in the related work.

3.3 Data filtering & evaluation

During the initial stage, duplicate entries identified were removed on the results obtained from the search string on all three databases. Following this, the remaining papers underwent two phases of screening and evaluation. In both phases, the papers were organized into two categories: *Accepted* and *Rejected*. Only the *Accepted* papers after the second phase were retained for final analysis and discussion.

In the first phase, the screening was conducted using the title and abstract to assess the relevance of each paper to the research objectives. The decision was guided by the extent to which the paper's content addresses the user interactions, use of AI or non-AI techniques, and adaptive immersions, etc., based on our research objectives and adhered to the inclusion and exclusion criteria. Our search string ensured that the intelligent, immersive XR environments are AI-enhanced; however, the user interaction component was not constrained to be based solely on AI.

In the second phase, all papers from the *Accepted* group of the first round were re-examined, this time using the full text for evaluation. This review was conducted collaboratively by the authors, where each author reviewed a subset of the papers. Each paper was evaluated to answer our defined RQs, and classified as *Accepted* for the final review and *Rejected* based on whether the paper's content addressed the research questions.

Table 4: Country-wise percentages of relevant published articles based on the extracted data in our SLR.

Country	Publications (%)
USA	19.46
India	16.77
China	12.80
Japan	4.42
Germany	3.12
France	2.61
UK	2.42
Australia	2.15
Egypt	1.72
Greece	1.38
Italy	1.17
Spain	0.87

At the conclusion of this process, only the finally accepted papers were retained for inclusion in the review. From an initial pool of 982 articles retrieved using the search strategy, 272 duplicates were eliminated, leaving 710 unique papers. After the first round of screening, 141 papers remained, and following the second round, this number was further refined to 93 papers. These 93 publications formed the final set for analysis and to address the defined RQs, as analyzed in detail in the following sections.

4 Results

This section systematically extracts the information from the finalized set of articles to answer RQ1, RQ2, and RQ3.

4.1 Publication information

The 93 selected papers were categorized in terms of publication year and country to show some basic information of our selected papers. Similar to the findings on smart immersive XR-based environments by Huang et al. [4], the number of publications in this field has rapidly been on the rise since the past few years (Fig. 3). The selected publications involve contributions from researchers representing 34 different countries and regions. To determine the distribution, we followed the scoring approach outlined in [4]. Each paper was assigned a total score of one. When all authors were affiliated with the same country or region, that entity received the full score. In cases where authors were spread across multiple countries or regions, the score was divided proportionally according to the formula "M/N," where N denotes the total number of authors and M represents the number of authors from a given country or region. Based on these calculations, the top five countries are: the USA, India, China, Japan, and Germany; see Tab. 4 for a list of the top ten countries with their percentage share of published papers.

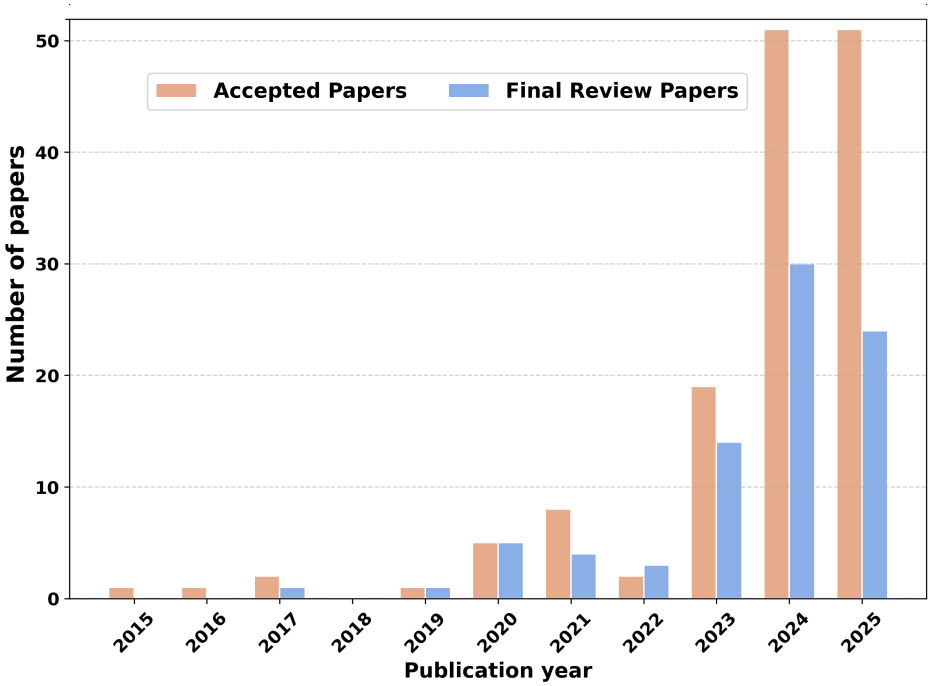


Fig. 3: Total number of publications received by executing our search query vs. those qualifying after the inclusion/exclusion criteria from 2015 to 2025.

4.2 Main applications, purposes, & modalities

We begin by answering RQ1, wherein the main applications, purposes, and modalities of user interaction in AI-enabled immersive environments are examined. The analysis of the selected studies reveals that AI-enhanced and human-centered immersive XR environments are applied across a wide spectrum of domains, each with distinct purposes and interaction modalities. These can be broadly categorized into education and training, healthcare and therapy, cultural heritage and creative industries, entertainment and gaming, industrial collaboration and design, retail and e-commerce, metaverse and social XR, as well as analytics and system-level frameworks. In the following, we first outline the application domains and purposes of interaction in AI-enhanced, human-centered immersive environments in Sec. 4.2.1, followed by the interaction modalities adopted in Sec. 4.2.2 (see Fig. 4, which represents the use of XR in different application domains).

4.2.1 Application domains & purposes

Education: Education represents the largest application domain for AI-enhanced immersive XR environments. Studies have targeted general educational enhancement, including analysis of student workload [8–10], immersive language learning [10–13], and personalization of VR experiences tailored to learner traits [14]. XR systems also

support special educational needs, including dyslexia interventions [15] and autism spectrum disorder training [16]. Collaborative learning, prosocial skill development, and adaptive engagement have been explored through interactive storytelling, role-playing mechanics [17], and multimodal fusion approaches [13, 14, 18–25].

Immersive XR has been applied for confidence assessment [26], procedural medical training [27, 28], science, technology, engineering and mathematics (STEM) and vocational training [18–20, 29–31], and programming comprehension with electroencephalography (EEG)-based cognitive engagement [21]. Collaborative embodied AR-based learning [32], situated and personalized AR learning [33], and job interview preparation [34] highlight the versatility of XR. Other works examine learner motivation, retention, and adaptive tutoring [22], as well as large-scale integration of AI speech robots for immersive education [35]. Additional studies explore engagement patterns and personalized interventions [18, 19, 30, 31, 36–39], demonstrating the capacity of XR to support both cognitive and affective learning outcomes.

In [40–42], the growing role of adaptive, multisensory interaction in XR-based education is highlighted. Their studies show that personalized immersion, real-time feedback, and haptic engagement enhance learner motivation, understanding, and embodiment, underscoring the value of intelligent, user-centered interaction in immersive learning environments. [43] present a recommender-based approach for personalized learning in immersive environments, where adaptive content delivery and interaction strategies are tailored to user preferences and progress.

Healthcare: Healthcare XR applications emphasize medical training, therapeutic interventions, and mental health support. VR and AR simulators support hands-on skill acquisition [27], adaptive clinical education [28], and phobia detection or stress behavior analysis [44, 45]. [46] explore intelligent interaction design for home environments aimed at supporting individuals showing early symptoms of Alzheimer's disease, while psychotherapy and rehabilitation benefit from immersive symbolic VR interactions and emotion-aware systems in [47, 48]. Emotionally adaptive XR is applied for telemedicine and empathetic communication [49, 50], and studies have also focused on anxiety reduction, hypnotherapy, bereavement support, and cognitive training [51].

Industry & collaboration: XR supports collaborative industrial work, training, and decision-making. Mixed reality enables remote collaboration [52, 53], holographic telepresence [53], Industry 5.0 workflows [54], human-in-the-loop cyber-physical monitoring [55], Social metaverse platform for multiuser interaction in XR environment [56]and human–drone training [57]. XR is also applied in interior design and space optimization [58], adaptive 3D space allocation [59], collaborative ideation [60], and robotics code generation or operator training [61, 62].

Cultural heritage & tourism: XR applications in cultural heritage focus on preservation, storytelling, and interactive museum experiences. Studies explore virtual kite-making [63], museum exploration [64, 65], textile heritage engagement [65], adaptive scene retargeting for artworks [66], and immersive cultural storytelling via VR museums [65]. XR has also been applied to VR concerts [67], virtual tourism [68], and emotion-aware adaptation in cultural experiences [33, 48, 66], demonstrating how immersive systems can enhance both engagement and learning in heritage contexts. [69] present a smart system designed to assist users in navigating and exploring

campus environments efficiently. [25] explore genAI for adaptive learning in XR environments, focusing on how AI-generated content and scenarios can personalize user experiences in real time.

Storytelling & entertainment: Entertainment-focused XR aims to enhance immersion, engagement, and replayability. Adaptive storytelling with AI-generated prompts [70], emotionally rich non-player character (NPC) dialogues [71], procedural level generation [72], neurofeedback for gameplay adaptation [73], prosocial skill development in children through digitally mediated collaborative role-playing [17], haptic feedback design to enhance user experience in game VR applications [74], and large-scale metaverse gaming with emotion-aware interfaces [75] illustrate the breadth of applications. Crowd simulation with heterogeneous personalities improves realism [76], while additional studies highlight interactive environments, haptic feedback, and multimodal fusion for immersive narrative experiences [60, 71, 77–81]. [34]integrated generative and discriminative AI to create immersive, interactive interview practice environments, similarly [67]enhance user engagement in VR live concerts by refining virtual audience avatar motions.

Visualization & data: XR applications centered on visualization and data emphasize how immersive environments enhance understanding, analytics, and collaboration across educational, industrial, and commercial contexts. Studies such as [9] and [82] showcase XR's potential in language education and hands-on interaction, while [59] introduces adaptive 3D spaces that dynamically respond to multi-user activities. Educational and industrial applications in [83], [84], and [54] integrate AR and knowledge graph-driven XR systems to facilitate data visualization, decision-making, and human–machine collaboration. Research by [55] focuses on human-in-the-loop XR systems that support real-time monitoring and adaptive control, whereas [85] explores immersive analytics through 3D point cloud interaction. User-centric modeling and attention prediction in VR are addressed by [86] and [87], improving performance and comfort in immersive environments. Meanwhile, [75] examines emotion-aware personalization in game-based XR systems. Finally, commercial and retail applications in [88] and [89] demonstrate how AR and MR enable interactive shopping and visualization experiences that merge digital content with real-world contexts.

Social XR & communication: In social and collaborative XR, researchers emphasize enhancing human connection, empathy, and communication across immersive environments. [53] explores holographic telepresence for multi-user collaboration, while [10] demonstrates how XR and AI can support second-language learning through immersive, task-based environments. The integration of multimodal, human-like interaction is advanced by [90], who enable personalized intelligent virtual agents capable of speech, visual expression, and gesture recognition, supporting natural communication in social XR. Building upon this, [91] applies immersive VR to grief therapy and emotional support, showing the social depth achievable through generative XR storytelling. Physiological and affective dimensions are also explored by [49] and [92], who integrate affective computing and adaptive notifications to improve user comfort and empathy in XR communication. Together, these studies illustrate the emergence of Social XR as a multidimensional medium for expressive, emotion-aware, and collaborative interaction. In the text that follows, the term "affective" will be used to

describe systems that respond to or detect emotional states, which involve affective computing, emotion-aware interfaces, or biometric feedback.

Technical & systems: Technical and system-level XR research focuses on the integration of intelligence, automation, and adaptability across engineering, manufacturing, and computing domains. In [21], immersive environments are leveraged for programming education, improving conceptual understanding and engagement through interactive visualization. [61] extends XR into Industry 4.0 applications, combining AI-based code generation and VR simulations to automate and optimize manufacturing processes. Complementing these, [71] explores emotionally intelligent NPCs and interactive storytelling in immersive games and simulations, while [93] proposes an enhanced multimodal XR framework that adapts to user behavior, cognitive load, and environmental variability.

Discussion: AI-enhanced XR spans diverse domains, with education and healthcare as dominant applications, followed by cultural heritage, entertainment, industrial collaboration, retail, social XR, and system-level analytics. These applications range from knowledge acquisition, training, and therapy to cultural engagement, entertainment, and decision support. Across domains, XR leverages adaptive, human-centered design to optimize engagement, presence, and effectiveness, while addressing both cognitive and affective needs.

4.2.2 Interaction modalities

The reviewed studies employed a broad range of interaction modalities, often combining multiple input and output channels to enable naturalistic, adaptive, and immersive user experiences. Interaction modalities in AI-enhanced immersive environments span a wide spectrum of sensory and cognitive channels, serving as the primary means through which users engage with virtual content. Based on the reviewed literature, these modalities can be grouped into six main categories: visual and gestural interaction, speech and audio-based interaction, multimodal fusion and adaptive feedback, haptics and tactile interfaces, physiological and affective sensing, and system-level/telepresence interaction. Each category reflects distinct design priorities and domain-specific affordances, revealing how different application contexts shape the choice and implementation of interaction techniques. Tab. 5 illustrates the diverse interaction modalities employed across various application domains for user interaction in AI-enhanced immersive environments. Each identified application domain, education, healthcare, industry/collaboration, cultural heritage/tourism, storytelling/entertainment, visualization/data, social XR/communication, and technical/systems, utilizes one or more of these modalities to enhance user engagement and interactivity. The references listed alongside each modality indicate the studies or implementations that leverage these interaction methods.

Visual & gestural interaction: Visual immersion and gesture-based input remain the most prevalent modalities across XR application domains, forming the foundational layer of interaction. In education, VR and AR headsets combined with hand gestures, gaze tracking, and controller input support learning tasks, spatial navigation, and embodied cognition [8, 10, 11, 13, 16, 18, 19, 31–33, 35, 37–42]. In healthcare, visual feedback facilitates phobia detection and therapeutic engagement

Table 5: Interaction modalities employed in various domains for user interaction in AI-enhanced immersive environments.

Application Domain	Interaction Modalities	References
Education	Visual & gestural	[8, 10, 11, 13, 16, 18, 19, 31–33, 35, 37–42]
	Speech & audio	[12, 18, 20, 28, 30, 36, 41, 43]
	Physiological & affective	[21, 26, 40]
	Multimodal fusion	[13, 14, 18–25]
Healthcare	Visual & gestural	[51]
	Haptics	[47]
	Physiological & affective	[45, 49–51]
	Multimodal fusion	[46, 49, 50]
Industry & Collaboration	Visual & gestural	[52, 53, 57, 62]
	Haptics	[57]
	System-level & telepresence	[52, 53, 56, 62]
	Multimodal fusion	[52]
Cultural Heritage & tourism	Visual & aestural	[25, 64, 66, 68, 69]
	Speech & audio	[48, 68]
	Multimodal fusion	[64]
Storytelling & entertainment	Visual & gestural	[17, 34, 67, 70, 76, 77, 79]
	Speech & audio	[34, 60]
	Haptics	[74]
	Physiological & affective	[76, 80]
	Multimodal fusion	[81]
Visualization & Data	Visual & gestural	[9, 54, 55, 59, 82–88]
	Multimodal fusion	[75, 88, 89]
Social XR & Communication	Visual & gestural	[10, 53, 90, 91]
	Speech & audio	[10, 90]
	Physiological & affective	[49, 92]
	System-level	[53, 91]
	Multimodal fusion	[90]
Technical & Systems	Visual & gestural	[21]
	Physiological & affective	[21]
	System-level	[61]
	Multimodal fusion	[71, 93]

[51]. Industrial and collaborative systems rely on gesture tracking and spatial overlays for task coordination and remote assistance [52, 53, 57, 62]. Similarly, cultural heritage and tourism applications leverage visual storytelling and gesture-based exploration of virtual artifacts [25, 64, 66, 68, 69], while entertainment, storytelling, and gaming environments emphasize immersive navigation and object manipulation through visual cues and gestures [17, 34, 67, 70, 76, 77, 79]. Visualization and data analytics platforms employ sophisticated gaze- and gesture-based interaction for adaptive feedback and exploration [9, 54, 55, 59, 82–88]. Social XR environments integrate visual avatars and gesture recognition for communication [10, 53, 90, 91], and technical systems apply gesture-based control for simulation and training [21].

Speech & audio interaction: Natural language and audio interfaces support conversational interaction, voice commands, and auditory feedback. In education, speech-based tutoring and dialogue-driven learning enhance engagement [12, 18, 20, 28, 30, 36, 41, 43]. Cultural heritage applications use voice-guided tours and audio storytelling for immersive narrative experiences [48, 68]. Entertainment platforms incorporate speech for NPC dialogue, music education, and role-play [34, 60], while social XR environments facilitate voice-based communication and avatar interaction [10, 90].

Haptic: Haptic feedback provides tactile cues that enhance realism and embodiment, improving the sense of presence in immersive systems. Healthcare applications leverage haptic gloves and hand-tracking for therapeutic interventions [47], while industrial systems integrate tactile feedback into collaborative robotics and drone control [57]. Entertainment platforms employ haptic augmentation for immersive gameplay [74].

Physiological & affective: XR systems integrate physiological sensing to adapt experiences based on cognitive and emotional states. In education, EEG-based brain-computer interfaces (BCIs) and attention modeling tailor learning experiences [21, 26, 40]. Healthcare platforms monitor stress, emotion, and cognitive load using facial expressions, EEG, and sentiment analysis [45, 49–51]. Entertainment XR employs affective sensing for emotion-aware content delivery [76, 80], and social XR environments integrate physiological signals for avatar adaptation and real-time feedback [49, 92]. Some technical systems also leverage cognitive modeling, for instance, for simulation control [21].

Multimodal fusion: Multimodal fusion combines visual, auditory, gestural, and physiological inputs into adaptive frameworks. In education, multimodal systems integrate visual, auditory, and affective cues to support personalized learning and diverse learning styles [13, 14, 18–25]. Healthcare platforms combine multimodal feedback for caregiving, emotion recognition, and adaptive notifications [46, 49, 50]. Industrial XR uses multimodal fusion to improve collaborative tasks and robotic control [52]. Cultural heritage applications combine gaze, gesture, and speech for immersive storytelling [64]. Entertainment employs adaptive multimodal avatars and visuals for creativity and engagement [81]. Visualization platforms integrate gaze, speech, and gesture to enable intuitive data exploration [75, 88, 89]. Social XR environments use multimodal inputs for avatar interaction and real-time adaptation [90], while technical systems employ context-aware fusion for simulation and control [71, 93].

System-level & telepresence interaction: At the infrastructural level, XR supports telepresence, multi-user collaboration, and hybrid interfaces. Industrial applications employ MR overlays, holographic communication, and robotic feedback for remote operation and training [52, 53, 56, 62]. Social XR platforms facilitate avatar-based communication and adaptive rendering for immersive social experiences [53, 91]. Technical systems integrate recommender-driven interfaces and system-level adaptation for simulation and control [61].

Others: Certain other minor categories exist as interaction modalities. For instance, environmental fusion aligns virtual content with physical surroundings. Red, green, blue, and depth (RGB-D) cameras, light detection and ranging (LiDAR) scanners, and simultaneous localization and mapping (SLAM)-based tracking support spatial registration, object anchoring, and context-aware interaction, underpinning MR and AR applications and facilitating gestural, haptic, and collaborative interactions [51, 84, 94].

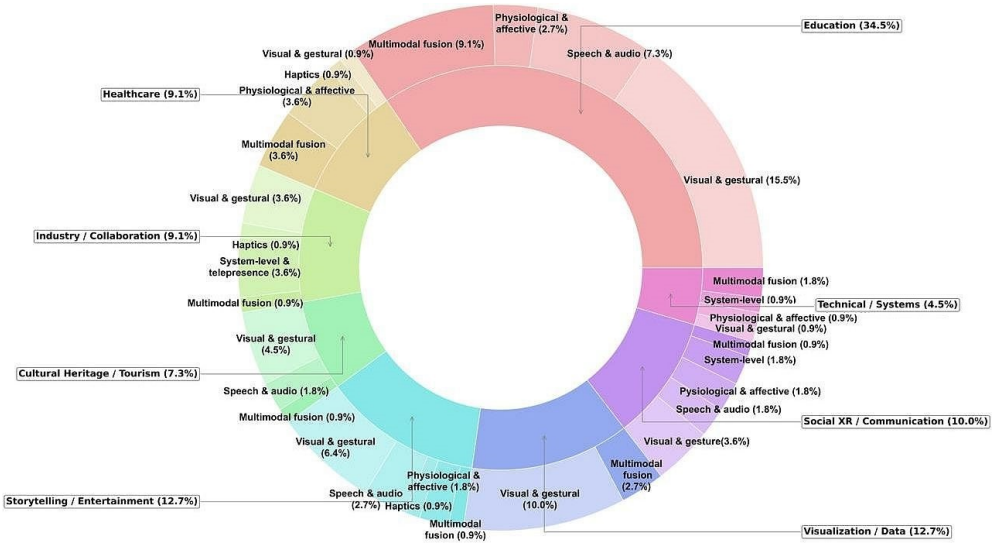


Fig. 4: XR applications across domains with corresponding interaction modalities (see also Tab. 5).

Discussion: In relation to Tab. 5, Fig. 4 presents XR application domains for various interaction techniques in AI-enabled smart environments. The inner ring once again shows the dominance of education & training (34.5%), followed by visualization & data (12.7%), storytelling & entertainment (12.7%), social XR & communication (10%), industrial design & collaboration (9.1%), healthcare & therapy (9.1%), cultural heritage & creative industries (7.5%), while technical & systems has the lowest share (4.5%). The most prominent modality is visual and gestural interaction, which accounts for 30.9% of the total, highlighting its widespread use in domains such as

education, entertainment, and data visualization. Speech and audio-based interaction represents 11.8%, commonly used across applications, such as, education, storytelling, and healthcare. Multimodal fusion, which integrates multiple input methods such as visual, audio, and haptic feedback, comprises 10.9%, indicating an increasing trend toward creating more immersive and realistic interaction experiences across various domains. Haptic interaction, although less dominant, accounts for 4.5%, and is particularly relevant in areas like healthcare and industrial training. Less frequent but significant modalities include physiological and affective interfaces, system-level interactions, and multisensory feedback, which appear more often in specialized domains such as healthcare, industry, and social XR. The mapping also highlights the prominence of a trend toward integrating multiple sensory inputs to create more immersive and intuitive user experiences.

4.3 Interaction hardware

Across domains, the reviewed studies employed a broad range of hardware and sensor setups (RQ2) to create immersive experiences and facilitate intuitive user interaction on AI-enhanced platforms. The hardware can be categorized into two different categories based on the type of hardware and the type of XR modality, respectively, as detailed below.

4.3.1 Hardware type

The hardware and sensor setups can be grouped into four major categories, as outlined below. These categories reflect the evolving landscape of XR technologies and their role in enabling immersive, adaptive, and multimodal interaction.

Immersive displays: HMDs remain the foundational interface for delivering immersive content, particularly in VR applications across education, healthcare, and entertainment [17, 54, 88]. Devices such as Oculus Quest, HTC Vive, and Meta Quest are frequently cited for their high-fidelity visual immersion. MR systems extend this paradigm by integrating holographic overlays through headsets like HoloLens and Magic Leap [10, 46], while AR applications leverage lightweight mobile platforms and AR glasses to superimpose digital content onto the physical world [32, 33]. These display technologies form the visual backbone of XR experiences, enabling spatial presence and contextual engagement.

Interaction devices: Physical interaction in XR is facilitated through a diverse array of controllers and sensors. Gesture-based controllers, haptic gloves, and robotic interfaces are widely adopted in VR and MR environments to support embodied interaction [26, 47, 57]. Motion capture rigs and treadmill systems further enhance realism in collaborative and training scenarios [62]. In AR contexts, interaction is often mediated through touchscreens, mobile dashboards, and camera-based input [77, 95]. These devices enable users to manipulate virtual elements with precision and responsiveness, bridging the gap between physical and digital actions.

Physiological & behavioral sensors: A growing trend in XR research is the integration of biosignal-based interfaces for adaptive interaction. EEG headsets, galvanic skin response (GSR) sensors, and electromyography (EMG) modules are

employed to monitor cognitive load, emotional states, and muscle activity, particularly in education and healthcare domains [87, 88]. Gaze tracking and facial expression recognition further extend this capability, allowing systems to personalize responses based on user affect and attention [49, 91]. These sensors underpin human-centered design by enabling real-time adaptation and affective computing.

Environmental & spatial sensors: To achieve spatial alignment and contextual awareness, XR systems incorporate RGB-D cameras, LiDAR scanners, and ambient sensors. These technologies are especially prevalent in MR and AR applications, where accurate mapping of the physical environment is critical for seamless integration of virtual content [9, 53, 84]. Such sensors support collaborative scenarios, object recognition, and dynamic scene understanding, enhancing the realism and utility of XR deployments.

4.3.2 XR modality

Another way to study the employed interaction hardware can be to classify it based on the underlying XR modality.

Virtual reality: Within the VR domain, a broad range of HMDs such as the Oculus Quest, Quest 2, Rift, HTC Vive, Vive Pro Eye, Valve Index, and smartphone-based VR headsets were commonly utilized to facilitate immersive experiences in educational, training, and collaborative contexts[8, 13, 14, 17, 21, 24, 26, 28, 37, 38, 40, 41, 44–48, 50, 51, 55, 56, 59–61, 64, 67, 68, 70, 74–76, 79–81, 84, 85, 87, 90, 92, 94]. Interaction in these systems was frequently supported by motion controllers, VR gloves, and gesture-based trackers for natural manipulation and locomotion, as seen in systems integrating leap motion sensors, eye tracking, and treadmill-based movement interfaces [42, 57, 86]. Several studies further incorporated biofeedback devices, such as EEG, EMG, and GSR sensors, to capture physiological responses for adaptive immersion [37, 86, 87]. Input and output modalities like microphones, webcams, haptic feedback units, robotic and drone controllers, and visual dashboards expanded user engagement and real-time monitoring [42, 48, 57]. High-performance computing systems powered by GPUs such as RTX 3080Ti, 3090Ti, and Tesla T4 enabled real-time rendering and AI-driven interactions [8, 59, 81]. Environment integration through Unity3D-based simulators, immersive rooms, and museum-like virtual spaces allowed for advanced analytics and educational experimentation [38, 40, 60, 80]. Lastly, several works highlighted the role of VR conferencing, avatars, and intelligent virtual agents for collaborative or social interactions [17, 47, 50], while physiological interfaces such as EEG-based headbands like Emotiv Epoc and NeuroSky were leveraged for neuroadaptive interaction design [37, 40, 86].

Augmented reality: In AR applications, head-mounted displays and AR glasses were increasingly employed to support contextualized learning, maintenance assistance, and interactive visualization [12, 15, 25, 32, 33, 83, 88, 89, 95]. Mobile devices such as smartphones and tablets equipped with ARCore or ARKit formed the most prevalent platform, enabling marker-based and SLAM-driven augmentation [49, 69, 77]. Touchscreens, microphones, and embedded cameras provided primary input-output interfaces for AR visualization, often enhanced with real-time dashboards

and overlay systems for improved interactivity [12, 15, 25, 33, 48, 95]. Additional interaction modalities were supported by external devices such as drone controllers and mobile UI interfaces [49, 57]. Tracking and detection relied primarily on built-in cameras and sensor fusion techniques for marker-based and markerless SLAM tracking [15, 77, 83].

Mixed reality: In MR environments, head-mounted displays such as Microsoft HoloLens, HoloLens 2, and the Meta Quest Pro XR headset were predominantly adopted to merge physical and virtual contexts in collaborative, educational, and industrial use cases [9, 10, 22, 31, 34, 52–54, 62, 65, 71, 91, 93]. Advanced sensors and tracking units, including gaze, LiDAR, RGB-D, and facial trackers, were commonly integrated to capture user and environmental data for adaptive rendering [10, 52, 53, 93]. Input-output systems featured holographic displays, telepresence rigs, and tablet interfaces, enabling seamless multimodal interaction [22, 34, 62, 71]. High-performance computing resources such as edge-cloud architectures and RTX 4090–based workstations were utilized for real-time MR rendering and AI-driven fusion [9, 31, 54]. Finally, several MR studies emphasized the role of adaptive and generative agents, such as storytelling avatars and multimodal sensing frameworks, in supporting context-aware collaboration [10, 65, 71, 93].

Discussion: Tab. 6 provides a comprehensive overview of the XR modalities identified in various publications, categorizing the associated hardware devices and interaction modalities used across VR, AR, and MR systems. It highlights how each XR type employs distinct technological components such as HMDs, sensors, controllers, computing systems, and integration or networking tools to facilitate immersive experiences. The extracted data emphasizes the diversity of devices ranging from VR headsets and motion controllers to AR-enabled smartphones and MR holographic displays, illustrating the hardware ecosystem that supports XR applications. Additionally, it includes references for each category, linking the listed technologies to relevant studies for traceability.

Connected to this, Fig. 5 visualizes the distribution of XR research across three primary categories: VR, AR, and MR, based on the proportion of publications focusing on specific technical components. The outer ring segments XR into VR (54.5%), AR (20.5%), and MR (25.0%). The inner ring further breaks down each category into component-level research areas, such as HMDs, input/output interfaces, sensors & trackers, computing systems, and environmental integration. VR research is predominantly focused on HMDs (35.4%), followed by various subsystems like controllers, biofeedback, and conferencing systems.AR research is more evenly distributed, with a significant focus on input/output interfaces (5.3%), mobile devices (2.7%), and HMDs (8.0%). MR research shows a balanced interest across multiple components, notably HMDs, sensors & trackers, and environment & agents.

4.4 AI vs. non-AI interaction

Tab. 7 provides a comprehensive mapping of various AI types and their corresponding features across multiple application domains in immersive environments. It categorizes AI techniques such as ML, deep learning (DL), reinforcement learning (RL), generative AI (GenAI), and unspecified AI (UnspecAI) based on their usage within various

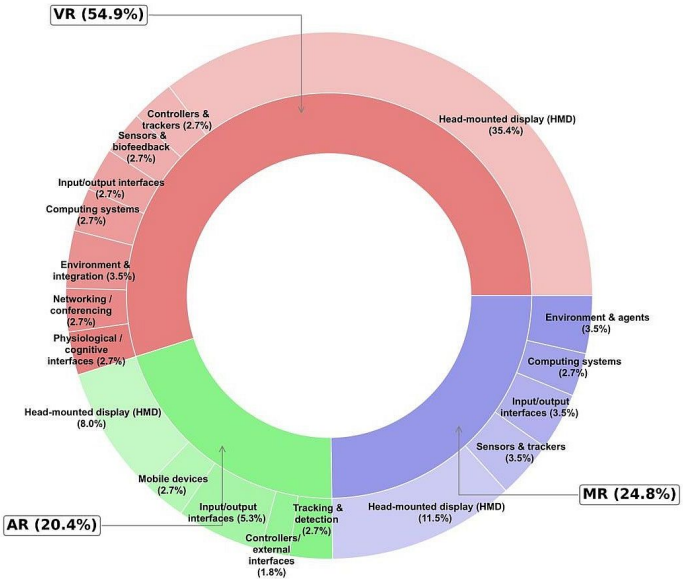


Fig. 5: Distribution of hardware devices and interaction modalities across XR applications (see also Tab. 6).

application domains. While AI is used in some form to build immersive intelligent systems, the (ₓ) symbol denotes instances where AI is not directly utilized for user interaction. This structured representation highlights how different AI paradigms are distributed across application areas, reflecting their relevance, adoption trends, and functional roles in immersive technologies.

4.4.1 Machine learning

Méndez et al. [14] use trait-based tailoring for personalization, mapping personality to content, and enabling adaptive tailoring of immersive learning experiences. [35] apply ML for experience modeling to personalize immersive educational experiences. [23] apply ML-driven interaction analysis to support responsive instructional scenarios in XR. In [19], ML models to assess learner performance and adapt feedback in immersive educational contexts are employed. MediaPipe and neural methods to support real-time user analysis in XR interactions are integrated in [38]. ML is used to estimate learner confidence and adapt content or feedback accordingly in [26], while [12] implement a customized ML-based automatic speech recognition (ASR) system for pronunciation assessment and live feedback in immersive language practice. [30] use ML pipelines to support sign-language recognition within AR/VR learning tasks. [42] leverage ML-driven sensing for embodied feedback in XR training.

Davies et al. [45] model stress and physiological responses with ML to personalize healthcare-oriented XR interventions. ML is applied to automate clinical workflows in XR-supported medical tasks in [44], to design intelligent ML modules for clinical decision support in immersive healthcare applications in [94], and to enhance patient

Table 6: XR modality employed in the identified publications with categorized hardware devices.

XR Type	Categorized Hardware Devices / Interaction Modalities	References
VR	**HMDs:** Oculus Quest / Quest 2 / Quest Pro, Oculus Rift, HTC Vive, HTC Vive Pro Eye, Meta Quest (all versions), Smartphone VR headsets, Valve Index	[8, 13, 14, 17, 21, 24, 26, 28, 37, 38, 40, 41, 44–48, 50, 51, 55, 56, 59–61, 64, 67, 68, 70, 74–76, 79–81, 84, 85, 87, 90, 92, 94]
	Controllers & Trackers: Motion controllers, Leap Motion, VR gloves, gesture gloves, hand tracker, eye tracker, Kat Walk mini treadmill, navigation controllers	[42, 57, 86]
	Sensors & Biofeedback: EEG headsets, GSR sensors, EMG sensors, facial sensors, head tracking sensors (3DOF), Tobii eye tracker	[37, 86, 87]
	Input/output Interfaces: Microphones, webcams (Logitech C270), VR haptics, robotic controller, drone controller, status board UI	[42, 48, 57]
	Computing Systems: PC with discrete GPU (GTX1080/1080Ti/RTX3080Ti), Nvidia Tesla T4/3090Ti, RTX GPUs	[8, 59, 81]
	Environment & Integration: Immersive rooms, museum simulators, mocap rigs, analytics dashboards, Unity 3D integration	[38, 40, 60, 80]
	Networking & Conferencing: VR conferencing tools, VR avatars, IVA cameras	[17, 47, 50]
	Physiological/Cognitive Interfaces: EEG headbands (Emotiv Epoc, NeuroSky)	[37, 40, 86]
AR	**HMDs:** AR glasses	[12, 15, 25, 32, 33, 83, 88, 89, 95]
	Mobile Devices: Smartphones/tablets (ARCore/ARKit-enabled), ARCore-enabled Android phones	[49, 69, 77]
	Input/Output Interfaces: Touchscreens, microphones, device cameras, webcams, AR dashboards, AR overlays	[12, 15, 25, 33, 48, 95]
	Controllers/External Interfaces: Mobile interfaces, drone controllers	[49, 57]
	Tracking & Detection: Cameras for marker detection / Simultaneous Localization and Mapping (SLAM)	[15, 77, 83]
MR	**HMDs:** HoloLens / HoloLens 2, MR headsets, Meta Quest Pro XR headset	[9, 10, 22, 31, 34, 52–54, 62, 65, 71, 91, 93]
	Sensors & Trackers: Gaze trackers, adaptive sensors, RGB-D sensors / cameras, LiDAR, facial trackers	[10, 52, 53, 93]
	Input/Output Interfaces: Holographic displays, tablets, RGB-D cameras, telepresence rigs, adaptive rigs	[22, 34, 62, 71]
	Computing Systems: Cloud-edge devices, high-end workstations (Ryzen 9 5950X, RTX 4090, 64 GB RAM)	[9, 31, 54]
	Environment & Agents: Storytelling agents, avatar systems, generative avatars, adaptive agents, multimodal sensing/fusion rigs	[10, 65, 71, 93]

monitoring and interaction in XR health systems in [50]. A recommender-style ML to surface relevant content in health-focused XR is adopted in [43]. [49] integrate ML-based context awareness to support collaborative XR medical scenarios, while [51] use ML in the design of XR health tools to optimize workflows and interfaces.

Zhang et al. [52] employ ML for adaptive instruction generation in industrial collaboration XR scenarios. ML techniques for adaptive industrial XR systems are applied in [56]. ML in cultural-heritage XR reconstructions is explored in [68], while it is used

Table 7: Mapping of different AI types and features across application domains in immersive environments ([x] indicates AI is not used for the user interaction).

Application Domain	ML	DL	RL	GenAI	UnspecAI
Education	[12, 14, 19, 23, 26, 30, 35, 38, 42], [37][x], [40][x], [41][x]	[21, 25, 35, 38, 42], [39][x]	[72]	[11, 13, 20, 25, 32, 36]	[55], [24][x], [8][x], [16][x]
Healthcare	[43–45, 49–51, 94]	[46, 49, 94]	-	[91]	[47]
Industry & Collaboration	[52, 56]	[53, 62]	[56, 57]	-	-
Cultural Heritage & Tourism	[48, 64, 66, 68]	[66]	-	[22, 65]	[69][x]
Storytelling & Entertainment	[76, 79, 80]	[34, 79]	-	[60, 70, 81]	[77], [17][x], [67][x], [74][x]
Visualization & data	[9, 84–86, 88, 89, 92]	[31, 59, 75, 82, 85–87, 96]	-	[88]	[33, 54, 83]
Social XR & Communication	-	[10, 49, 53, 90]	-	[28, 90]	-
Technical & Systems	[21, 93], [62][x]	-	[93]	[71], [61][x]	-

to enable advanced perception and interaction in cultural-tourism XR in [66]. Similarly, [64] use ML and fuzzy methods to support heritage-guided XR experiences, and [48] for evaluation of ML-assisted feasibility in heritage XR deployments. ML-based authoring tools for storytelling & entertainment XR are employed in [76], immersive narrative adaptation in [79], and to the end of immersive entertainment interactions in [80].

Senthil and Prabha [9] leverage computer-vision and edge-ML for real-time interactive visualization in XR. ML is utilized to scale large-model-driven visual analytics for XR data tasks in [88], immersive information retrieval and data representation in [89], and interactive ML or IML workflow visualization in [85]. Several domain-specific examples of ML use exist. This includes ML for medical in XR [86], "Goldilocks" strategies for adaptive visualization in XR [92], knowledge-graph/metadata (MetaKG) integration in XR data pipelines [84], user modelling and adaptive control [21], improved system-level adaptation and policy tuning in XR environments [93], and ML-informed system design patterns for XR technical systems [62].

4.4.2 Deep learning

In [21] and [35] employ deep learning methods (Convolutional neural network (CNN), transformer variants) for perception and experience modeling in educational XR . Deep networks for embodied sensing and feedback generation in training scenarios are applied in [42]. [25] investigate generative deep models to synthesize learning content and affordances in XR. Deep neural architectures for real-time pose and gesture recognition in immersive interfaces are used in [38]. [39] examine non-user-facing deep-input techniques for interaction logging and offline analysis. In the field of healthcare, DL for intelligent XR tools in clinical workflows are used in [46]. Deep networks for automated analysis and intelligence in healthcare XR modules are applied in [49, 94]. Deep models for context-aware industrial XR design are explored in [53, 62]. [34] and [79] combine deep generative and discriminative models for storytelling and immersive narrative adaptation. Deep architectures to improve perception, reconstruction and XR rendering for cultural-tourism and visualization tasks are used in [31, 66]

. [59, 75, 82, 87, 96] document domain-specific deep methods to drive interactive XR visualization and rendering. [85, 86] demonstrate deep-learning-enabled Immersive Machine Learning (IML) /visualization hybrids for domain analytics in XR. In social XR, DL enabled realistic avatar interactions, emotion recognition, and context-aware immersive experiences, enhancing the sense of presence and social connectedness [10, 49, 53, 90].

4.4.3 Reinforcement learning

Fewer user interaction techniques leverage RL when compared to ML and DL. [72] explore RL for virtual training scenarios, where policies guide adaptive tutoring behaviors in XR. RL is applied for action control in drone-archery and interactive physical-XR tasks, demonstrating the technique in embodied scenarios in [57]. RL is used to enable adaptive industrial controllers in collaborative XR workflows in [56]. [93] combine RL for policy optimization to improve system responsiveness and user-centric control in technical XR systems.

4.4.4 Generative AI

Generative approaches to synthesize immersive content and narrative scaffolds for XR education are leveraged in [13, 20, 32]. [11] examine how large language models (LLMs)-driven architecture can author contextual guidance and dialogues in learning XR. [25, 36] apply generative models to create interactive content and multimodal assets for immersive storytelling and training. LLM/vision-language Model (VLM) prompt pipelines for context-aware assistance and interview/simulation scenarios in XR are investigated in [81, 88]. [60, 70] consider generative dialog and control for immersive narratives and guided experiences. [91] applies generative approaches for intelligent healthcare narration and support tools in XR clinical settings. [22, 65] use generative XR tools to create curiosity-driven heritage tours and enhanced tourism experiences. [28] and [90] highlight LLM-based context awareness and social XR communication scaffolds. [61, 71] examine humanizing generative processes and system-level generative components that are sometimes non-interactive in the user loop.

4.4.5 Unspecified AI

Certain other works leverage AI for user interaction in immersive AI-based environments, without specifying the nature of AI. [55] present frameworks to enhance collaborative immersive learning with AI-supported scaffolding, without detailing a single algorithmic family. Immersive workflows in [8, 24] provide system-level frameworks where AI is not directly used for user interaction. Alimanova et al. [16] discuss developing immersive experiences with system-centered AI components rather than user-facing modules . [47] describe co-design frameworks that include AI as a background orchestrator in healthcare XR [47]. Divya's Aroute is listed as non-user-facing and focuses on route and AR frameworks rather than explicit AI-user loops [69]. [17, 77] provide comparative and authoring frameworks for XR storytelling where AI roles are not explicitly specified. [33, 54, 83] discuss AR/ XR system design and

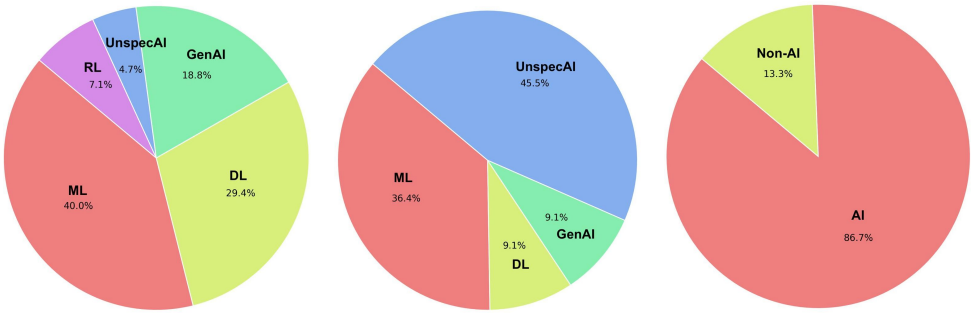
authoring environments where AI is present in the pipeline but not always defined to the user-facing interaction level.

4.4.6 Non-interactive AI

Several studies included in Tab. 7 apply AI-related frameworks or analytics indirectly, without engaging the user through immersive AI-driven interaction. In the education domain, [37, 40, 41] focus on AI-supported design and analytics frameworks for learning systems but do not employ AI for direct XR interaction. Similarly, [39] explore *GInput* for gesture input analysis as a data-collection mechanism rather than a user-facing AI module. Immersive education and training systems with embedded AI logic but without adaptive, AI-mediated user interfaces are described in [8, 16, 24]. AI-based optimizations for system design in industrial XR applications are included in [62], where the intelligence functions are system-centered and not interactive. In cultural heritage, the ARoute project in [69] focuses on route planning and AR path management with limited AI-user engagement. Likewise, [17, 67, 74] develop storytelling and entertainment frameworks where AI operates as a backend process, not an immersive co-actor . Finally, [61] explore generative system-level architectures intended to humanize design workflows but without involving direct AI-user interactions.

Discussion: Overall, the findings for RQ3 reveal a continuum. Non-AI static systems rely on predefined scenarios, role-playing, or haptics, offering immersion but limited personalization. AI-driven adaptive systems enable real-time personalization via ML, RL, conversational AI, and emotion recognition. Hybrid systems strategically combine AI personalization with structured, non-AI interaction mechanics to balance control, usability, and adaptivity. This analysis highlights that while non-AI techniques remain prevalent for ensuring reliability and realism, AI-driven adaptivity increasingly enhances personalization and human-centered interaction in XR. Hybrid systems that integrate both paradigms appear especially promising for future XR applications, as they allow personalization while maintaining interpretability and structured user engagement; Fig. 6 represents the share of AI vs non-AI approaches in the reviewed papers.

Fig. 6 presents a comparative analysis of AI involvement in XR interaction. Fig. 6a illustrates the distribution of AI subfields specifically used to enable XR interaction, with ML (40.0%), DL (29.4%), GenAI (18.8%), RL (7.1%), and UnspecAI (4.7%) contributing to various adaptive and immersive functionalities. In contrast, Fig. 6b represents cases where AI was not used for XR interaction, showing a dominance of UnspecAI references (45.5%) despite the absence of direct AI-driven adaptivity, followed by ML (36.4%), and smaller shares of DL and GenAI (9.1% each) with no use of RL. Fig. 6c simplifies the comparison, revealing that 86.7% of XR studies incorporated AI techniques for interaction, while 13.3% relied on non-AI approaches. This visual summary underscores the growing prevalence of AI in shaping personalized, multimodal XR experiences, while also highlighting the continued use of traditional, static interaction methods in a significant portion of the research.



(a) (b) (c)

Fig. 6: Distribution of AI-enhanced immersive systems by (a) AI subfields used in XR interaction, (b) AI not used for interaction, and (c) overall AI vs. non-AI usage, see also Tab. 7.

4.5 Limitations

Despite significant progress in AI-driven XR systems, a comprehensive review of the literature reveals recurring and diverse limitations across current research. These constraints can be organized into several interrelated clusters that collectively highlight the maturity gaps, implementation barriers, and ethical considerations of current AI-enhanced XR systems.

4.5.1 Lack of personalization & adaptive intelligence

Many studies continue to demonstrate static or limited personalization in immersive XR systems to the end of user interactions. Adaptive mechanisms often rely on simple rule-based or performance-driven feedback without capturing user-specific traits, learning pace, or cognitive differences. Systems such as [10, 14, 17, 30, 32, 40, 41, 44, 63, 64, 68, 74] highlight that adaptivity is typically pre-scripted or coarse-grained, limiting the capacity to model complex affective or behavioral responses. This often results in experiences that are uniform across users, reducing immersion and engagement. Additionally, emotional or cognitive adaptation remains underexplored, with most frameworks neglecting long-term personalization across learning or rehabilitation contexts.

4.5.2 Technical, computational & hardware constraints

A broad subset of work underscores persistent technical bottlenecks, including latency, real-time synchronization, and resource intensity of AI models. Deep or generative XR architectures [13, 25, 31, 52, 53, 57, 60, 61, 80, 97] face challenges such as frame-rate instability, limited rendering power, and high computational demand for multimodal fusion. XR systems that integrate blockchain or decentralized architectures further encounter scalability and transaction overhead issues [29, 59, 68]. Hardware-related constraints like occlusion, motion tracking errors, and calibration complexities impede

fluid interaction, while limited optimization for lightweight devices continues to restrict portability.

4.5.3 Evaluation, scalability & dataset limitations

Numerous studies rely on small-scale prototypes or controlled laboratory trials, making it difficult to assess scalability and real-world applicability. Work such as [11, 19, 22, 34, 54, 59, 70, 76, 88, 95] are constrained by limited participant pools or specific device dependencies, which restrict external validity. Furthermore, datasets are often domain- or language-specific, impeding generalization across educational, cultural, or industrial settings. The lack of large, diverse, multimodal datasets reduces model robustness. Studies like [20, 41, 50] also emphasize infrastructural and pedagogical barriers, such as limited teacher training, insufficient computational infrastructure, and the absence of standard evaluation frameworks for XR performance metrics.

4.5.4 Modality-specific & interaction limitations

Interaction fidelity remains a major challenge in XR systems. Voice-based modalities often exhibit low recognition accuracy or latency issues under varying acoustic and linguistic conditions [12, 18, 23, 25, 95]. Gesture and vision-based modalities are hindered by occlusion, lighting sensitivity, and limited gesture libraries [38, 82]. Multimodal systems, although conceptually powerful, are frequently underdeveloped or poorly synchronized, with several studies relying solely on visual or textual cues [15, 36, 49, 64, 67, 98]. The absence of fine-grained integration between gaze, haptic, and vocal inputs constrains the realism of immersive feedback and social presence.

4.5.5 Ethical, privacy & inclusivity challenges

A critical dimension of limitation across the literature involves ethical and privacy concerns. Multi-sensor XR systems inherently gather sensitive user data, motion, speech, and biometrics, raising issues of consent and surveillance [18, 46, 94]. Bias in AI models leads to unequal representation and cultural stereotyping in avatars or adaptive responses [79, 94, 96]. Accessibility and inclusivity remain under-addressed, with few systems accommodating users with disabilities or non-dominant languages [10, 20, 50, 65]. Cost barriers persist for many XR devices, making equitable deployment across socioeconomic contexts difficult. This cluster also encompasses growing concerns regarding explainability and transparency of AI decisions in immersive contexts.

4.5.6 Contextual, usability & deployment barriers

Several studies note that while XR systems perform effectively in research settings, their transition to real-world deployment is limited by contextual mismatches and usability issues. Systems designed for education or training environments often require controlled conditions that are difficult to replicate in field scenarios [9, 27, 41, 45, 75]. Usability constraints include steep learning curves, fatigue due to long usage durations, and the absence of standardized user experience benchmarks. Environmental

variability, lighting, space constraints, or noise further hinder consistent interaction. As identified by [54, 65], these contextual limitations underscore the importance of design generalization and adaptive calibration across domains.

4.5.7 Algorithmic & model-specific limitations

At the algorithmic level, limitations often stem from the complexity, opacity, or resource intensiveness of AI models used within XR. Generative and reinforcement learning models demand substantial training data and computational resources, while lacking explainability [25, 61, 97]. Studies such as [85, 86, 99] also point out that interpretability and model transparency are underexplored in immersive contexts. Cross-domain transfer remains limited, with most models trained for narrow applications such as gesture recognition or speech feedback failing to generalize. Moreover, hybrid AI pipelines often suffer from pipeline fragmentation and latency during inference.

4.5.8 Pedagogical, cognitive & application-specific constraints

Finally, within education, training, and rehabilitation domains, XR studies reveal pedagogical gaps in integrating AI adaptivity into curriculum and cognitive design. Educational XR systems [17, 20, 29, 50] often overlook teacher-in-the-loop design and fail to balance automation with human agency. In therapy and rehabilitation contexts, limited longitudinal validation [8, 45, 72] hinders the establishment of sustained behavioral outcomes. Domain-specific XR systems for healthcare, cultural heritage, and industrial safety remain fragmented, lacking unified frameworks for ethical personalization, standard evaluation, or cross-context transfer.

In summary, while the reviewed studies advance AI-augmented immersion, they remain constrained by personalization gaps, scalability issues, technical bottlenecks, and ethical uncertainties. Addressing these limitations through multimodal adaptivity, scalable AI infrastructure, and human-centered design principles will be essential for realizing the full potential of next-generation XR ecosystems.

4.6 Opportunities

Building on the identified limitations, recent literature emphasizes several promising directions for advancing AI-enhanced XR toward greater adaptivity, inclusivity, and real-world scalability. These opportunities extend across multiple dimensions, ranging from algorithmic and interaction-level innovation to ethical, infrastructural, and interdisciplinary integration.

4.6.1 Real-time adaptive & emotion-aware interaction

Future systems are moving toward intelligent adaptivity that continuously adjusts to user behavior, affective states, and environmental feedback. Studies such as [14, 17, 40, 45, 68, 70, 75] highlight the potential of AI-driven personalization based on multimodal signals, eye gaze, voice, motion, and physiological data, to enable

dynamic scaffolding and emotion-aware feedback. Emerging frameworks propose predictive adaptation mechanisms that integrate real-time analytics with reinforcement learning, providing personalized trajectories in education, therapy, and industrial simulations. The incorporation of biosignals for workload and stress detection further enhances adaptive interaction and cognitive alignment between user and system.

4.6.2 Multimodal, collaborative & social XR interaction

Several works point toward richer, more immersive, and collaborative XR environments that integrate multimodal input channels. Combining gestures, haptics, gaze, and voice sensing can produce more natural and intuitive interfaces [8, 15, 49, 72]. Collaborative VR frameworks [26, 29, 55, 77, 88] explore multi-user adaptive learning, cooperative problem-solving, and decentralized communication through blockchain. These approaches envision shared environments where multiple participants interact synchronously, guided by adaptive AI tutors or avatars capable of responding contextually to group dynamics. Expanding such social XR models can enhance inclusivity, empathy training, and remote team collaboration.

4.6.3 Scalability, edge computing & cloud-driven expansion

One of the most promising frontiers lies in scaling XR systems through cloud and edge computing. Studies such as [31, 53, 56, 59, 62, 65] identify the convergence of 5G/6G connectivity, cloud AI processing, and Internet of Things (IoT) devices as key enablers of low-latency, high-fidelity XR applications. Offloading heavy AI inference to the edge or cloud can reduce device limitations and enable seamless personalization across platforms. This scalability pathway also supports large-scale educational and healthcare XR deployments, allowing real-time synchronization across devices and geographies while preserving computational efficiency and privacy.

4.6.4 Domain expansion & cross-disciplinary applications

The reviewed works suggest a strong movement toward expanding AI-enhanced XR into new domains such as healthcare, cultural heritage, industrial training, and cognitive rehabilitation. Examples include [23, 47, 65, 71, 91, 94, 98]. In these contexts, emotion-aware NPCs, empathy-based simulations, and domain-specific personalization are envisioned to enhance engagement and authenticity. In healthcare and therapy, adaptive XR can support patient monitoring, pain management, and neurorehabilitation. Cultural and heritage-focused XR applications leverage AI to enable interactive storytelling and cross-cultural education, creating opportunities for affective and experiential learning that bridge cognitive and emotional engagement.

4.6.5 Explainable, ethical & inclusive AI frameworks

Addressing the ethical and inclusivity challenges identified earlier, future research emphasizes explainability, fairness, and transparency in immersive systems. Studies such as [10, 18, 20, 50, 79, 94] advocate for the integration of ethical AI design principles that ensure data protection and equitable access. Explainable AI (XAI) is seen as

critical for maintaining user trust, particularly in high-stakes contexts like education and healthcare [85, 86, 99]. The development of culturally adaptive avatars, multilingual speech systems, and accessibility-oriented design frameworks promises more inclusive XR environments that cater to diverse learners, languages, and abilities.

4.6.6 Standardization, benchmarking & evaluation frameworks

Given the diversity of experimental approaches in existing literature, the establishment of standardized evaluation frameworks is essential. Studies such as [34, 41, 76**?**] propose structured validation metrics encompassing usability, immersion, emotional impact, and learning effectiveness. Future XR systems can benefit from open benchmarking datasets and shared validation protocols to enhance reproducibility and comparison across studies. Additionally, participatory evaluation, engaging educators, clinicians, and end-users in co-design, can ensure practical relevance and real-world viability.

4.6.7 Hybrid intelligence & autonomous adaptivity

Another emerging opportunity lies in combining symbolic reasoning with neural models to achieve hybrid intelligence in XR systems. Works such as [25, 52, 61, 97] propose integrating generative AI and rule-based logic for both content creation and adaptive control. Such approaches can support automated narrative generation, dynamic scene adaptation, and self-adjusting difficulty levels in interactive training or gaming environments. These systems promise scalable personalization while maintaining interpretability and control, balancing automation with human oversight.

4.6.8 Sustainable, cost-effective & accessible XR deployment

As cost and accessibility remain limiting factors, future research is focusing on lightweight, sustainable XR designs that minimize hardware demands. Efforts such as [20, 50, 54, 65] highlight cloud-integrated, mobile-first XR architectures that reduce entry barriers. Moreover, modular design and open-source ecosystems can promote broader adoption in education and developing regions. Integrating AI-driven optimization can also lower energy consumption and enable scalable XR experiences across affordable consumer-grade devices.

In summary, the reviewed opportunities reveal a vibrant trajectory for AI-enhanced XR research. Integrating real-time adaptivity, multimodal fusion, scalable architectures, domain diversification, and ethical frameworks will be pivotal for achieving robust, inclusive, and context-aware XR ecosystems. Future work must focus on unifying these directions through cross-disciplinary collaboration, standardized evaluation, and responsible AI practices to fully realize the promise of human-centered immersive intelligence.

5 Conclusion

This SLR synthesized research on user interaction in human-centered or AI-enhanced immersive XR environments across education, healthcare, training, entertainment, and

collaborative domains. By examining relevant published studies, we investigated four research questions: application domains and interaction modalities (RQ1), hardware interfaces and sensor technologies (RQ2), the role of AI and non-AI techniques in personalization (RQ3), and current limitations and opportunities (RQ4).

Our analysis shows that XR interaction has evolved from static, gesture-based systems to adaptive, context-aware environments that integrate multimodal sensing and intelligent adaptation. AI methods, including ML, RL, and generative models, play a growing role in tailoring experiences, anticipating user needs, and enabling emotion-aware or collaborative interactions. At the same time, non-AI approaches, such as rule-based personalization, procedural modeling, and predefined multimodal mapping, remain valuable for lightweight, interpretable, and resource-constrained deployments. *It is also worth noting that whereas the current technologies have led to the development of several immersive XR-based AI-centered environments, the use of AI in interaction is still catching up.*

The review highlights that hardware and sensor ecosystems, ranging from HMDs and depth cameras to physiological and affective sensors, are key enablers of real-time adaptive interaction. However, issues of scalability, latency, data quality, and interoperability remain open challenges. Similarly, while XR applications demonstrate strong potential in enhancing learning outcomes, supporting therapy, or enabling immersive collaboration, many studies remain domain-specific and lack generalizable frameworks for personalization.

Looking forward, the field faces both challenges and opportunities. Limitations include fragmented evaluation methodologies, small sample sizes, limited longitudinal studies, and insufficient attention to ethical, inclusive, and privacy-preserving design. Yet, emerging opportunities span real-time adaptive guidance, multimodal fusion, scalable infrastructures, cross-domain deployment, and integration with explainable and responsible AI.

In conclusion, this SLR underscores that the future of XR interaction lies in designing intelligent, adaptive, and ethically responsible systems that are both technically robust and deeply human-centered. By bridging AI-enhanced adaptivity with inclusive interaction design, next-generation XR platforms can deliver personalized, accessible, and transformative experiences across diverse application domains.

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